

Neural-based Rendering and Application

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ABSTRACT

Rendering plays an important role in many fields such as virtual reality and film, but the high dependence on computing sources and human experience hinders its application. With the development of deep learning, neural rendering has attracted much attention due to its impressive performance and efficiency than traditional rendering. In this paper, we mainly introduce two neural rendering works, one is rendering simulation and the other is image-based novel view rendering. Moreover, we also discuss the potential applications (i.e. data augmentation) based on the results of neural rendering, which has received little attention.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Neural Rendering, Point Cloud, Novel View Synthesis, Data Augmentation, Neural Network

ACM Reference Format:

Peng Dai. 2021. Neural-based Rendering and Application. In *Proceedings of the 29th ACM International Conference on Multimedia (MM '21)*, October 20–24, 2021, Virtual Event, China. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3474085.3481031>

1 INTRODUCTION

Rendering is an important part of computer graphics, which utilizes geometry, material, light, camera information to mimic the light transport in virtual 3D scenes and eventually synthesize images with different styles, such as cartoon and photo-realistic images. Furthermore, images in different camera views can be generated by changing the camera poses.

In the last few decades, rendering has been widely adopted in different areas (e.g. film, game, virtual reality and industrial design, etc.) and a large number of works, theories [13] and specific hardware were developed to support rendering techniques. However, there are still two main obstacles hindering the traditional rendering in practice. The first obstacle is the speed of rendering. Usually, rendering a high-quality image (physically based and noise-free) requires hours even days to accomplish the ray-tracing

process. The second obstacle is the complex manipulation of virtual models, which heavily relies on experienced artists and is also time-consuming. In addition to the image generation task, how to use rendering techniques to facilitate other down-stream tasks, such as semantic segmentation [11], is also worth exploring.

Recently, with the development of deep learning, rendering can be efficiently implemented using convolution neural networks (CNN), such methods are referred to as neural rendering [17]. In this paper, we mainly introduce two works of neural rendering, including rendering simulation and image-based novel view synthesis with the neural network. Specifically, the first work proposed PBR-Net [4] which takes essential rendering components as inputs to simulate the rendering process and generates photo-realistic images in almost real-time; and the second work directly synthesizes novel view images from point clouds (generated from images) with a multi-plane structure [5], which is able to handle occlusion relationships and generate temporal consistent videos. What's more, considering the huge demand for data in high-level tasks, such as semantic segmentation, a potential application of neural rendering is to provide data for downstream tasks which are currently in my research progress and will be discussed in this paper.

2 RELATED WORK

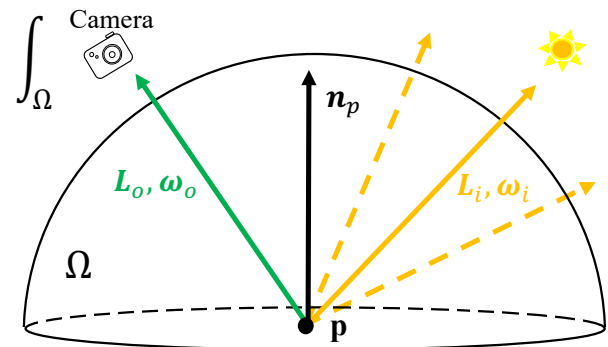


Figure 1: Rendering function.

2.1 Image Generation

One representative framework of image generation is the generative adversarial network (GAN) [2], which synthesizes photo-realistic images from random noise, sketch, and semantic maps [8, 9]. However, the image generation process is difficult to control. In order to analyze what has been learned by GAN and generate images with specific styles, Yang et.al [18] proposed to learn network representations for scene synthesis. In contrast to GAN, generating images from intrinsic components of rendering facilitate interpretability.

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MM '21, October 20–24, 2021, Virtual Event, China

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ACM ISBN 978-1-4503-8651-7/21/10...\$15.00

<https://doi.org/10.1145/3474085.3481031>

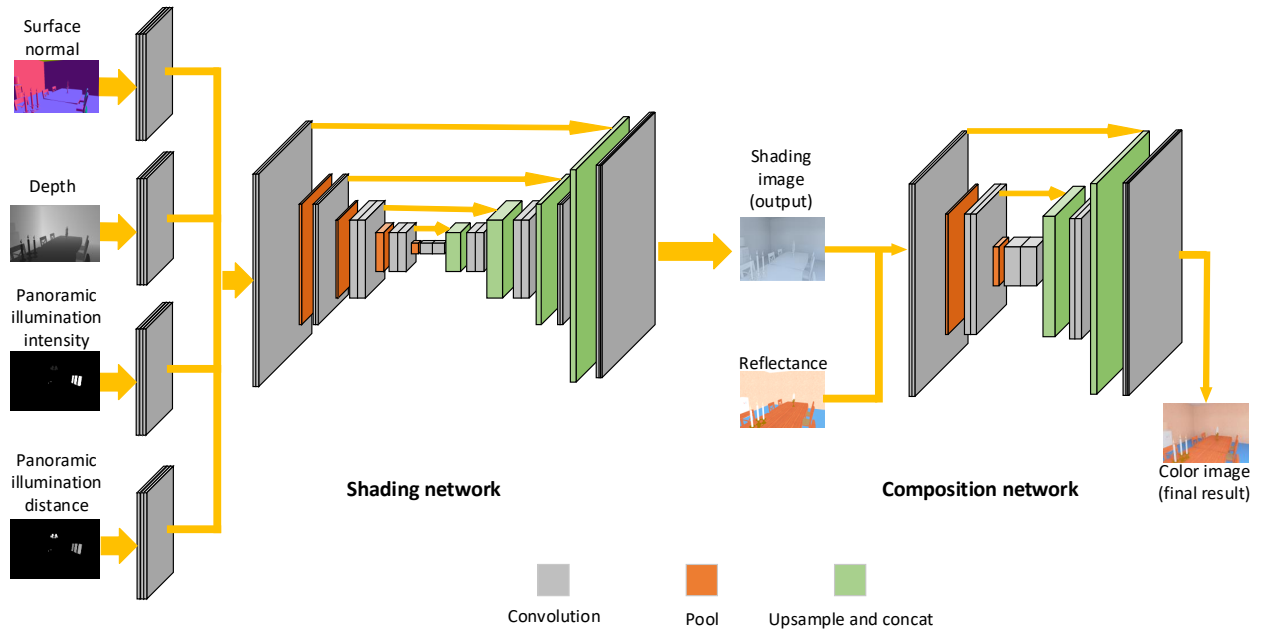


Figure 2: Rendering simulation. The PBR-Net imitates rendering process by providing the essential components required by traditional rendering.



Figure 3: Rendering result. The result generated by our method (PBR-Net) is closer to the ground truth.

2.2 Neural Rendering

Recently, using the neural network to implement rendering-related tasks are very popular. For example, Sun et.al. [15] utilize a convolution neural network to realize single image portrait relighting, and Mildenhall et.al. [12] proposed a novel framework named Nerf which successfully renders novel view images and inspires a series of following works [14, 20]. Compared with classic rendering, neural rendering has advantages in efficiency and concision.

2.3 Image Decomposition

Image decomposition is the inverse process of rendering, which decomposes images into different components, including shading, reflectance, roughness, illumination [6, 10, 19]. Next, image-based manipulations, such as image relighting, can be realized based on these components without complex rendering model construction.

2.4 Data Augmentation

Data augmentation is an efficient way to increase the diversity of training data and is widely used in deep learning to boost performance. Different from the traditional data augmentation strategies, such as random scale, rendering-based data augmentation is more physically reasonable. For instance, Halder et.al. [7] render rain streaks and Zhang et.al [21] render a massive amount of indoor images for augmentation, both of them achieve better performance than random augmentations.

3 PRELIMINARIES OF RENDERING

The following part introduces the background knowledge of classical rendering function, which is shown in Fig. 1 and can be mathematically formulated as

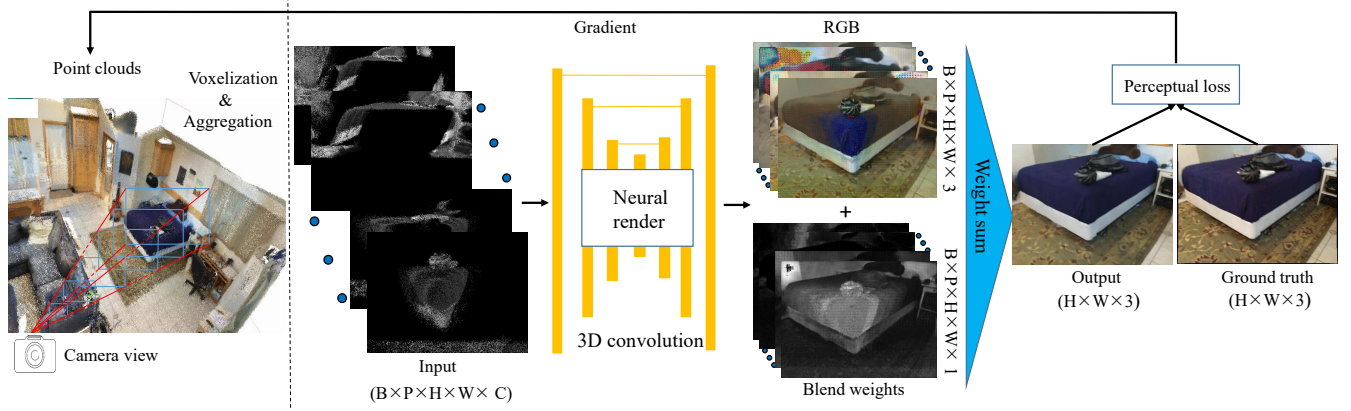


Figure 4: Novel view rendering. Our approach synthesizes images in different camera views based on the multi-plane structure.

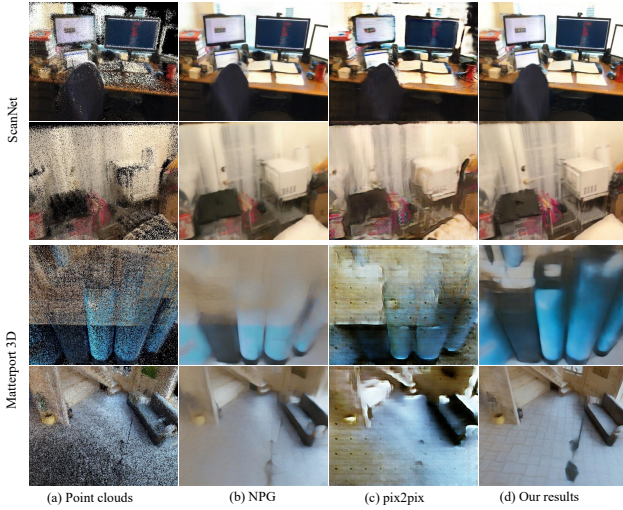


Figure 5: Novel view rendering results. Our approach correctly deals with occlusion relationship and generate high quality novel view images.

$$L_o(p, \omega_o) = \int_{\Omega} f(p, \omega_o, \omega_i) \odot L_i(p, \omega_i) (\omega_i \cdot n_p) d\omega_i, \quad (1)$$

where the rendering process can be represented as an upper hemisphere (Ω) integration. More specifically, p is a point in 3D space and n_p is the corresponding surface normal. The radiance L_o observed by camera from direction ω_o is affected by the incident radiance L_i from direction ω_i and the material f (e.g. bidirectional reflectance distribution function (BRDF)) which reflects the relationship between incident and outgoing lights.

4 RENDERING SIMULATION

The traditional rendering techniques using ray-tracing are highly explainable but computationally expensive, and the neural network for image generation is efficient but less of control. To combine the

advantages of these two approaches, we propose to use a neural network to simulate the rendering process.

4.1 Method

As shown in Fig. 2, the inputs consist of scene geometry (surface normal and depth), light information (panoramic light distance and intensity), material (reflectance/albedo), shading images, and the output is rendered RGB images.

Our design is inspired by the intrinsic image decomposition which decomposes RGB images into shading (more difficult to learn) and reflectance (not affected by light variation). Therefore, our approach learns shading with a larger network and then composes it with reflectance, which is more physically reasonable. Another reason for adopting shading images as intermediate supervision is because light effects, such as shadow, are more obvious in shading images. In other words, light effects are not distracted by reflectance. As for the backbone, we adopt the widely used U-Net structure with skip connections. (More details in the full paper [4])

4.2 Results

Time Cost. We evaluate our approach on an i7, 4 core CPU, and it takes 10.2s to render an image with 360×480 resolution, while about 3 minutes are required by traditional rendering using Mitsuba. Moreover, our method benefits from GPU, which takes 0.145 seconds to generate the output on an NVIDIA 1080Ti. In addition, the time used to extract rendering components can be ignored when compared with traditional rendering.

Qualitatively Comparison. In Fig. 3, the results generated by our approach are more photo-realistic and closer to the ground truth. Specifically, the reflection on the sofa and the shadow around the teapot and fruits on the table are more obvious.

5 NOVEL VIEW RENDERING

Given multiple frames captured in the same scene, how to render novel view images with fewer artifacts caused by the misalignment between images? Here, we proposed a multi-plane based approach to generate images with higher quality and temporal consistency.

5.1 Method

As shown in Fig. 4, the input of our method is point cloud constructed from multiple images. Different from other methods that project points to an image plane and select the closest one with z-buffer, our method proposed to project points to multiple planes and softly aggregate them for the following three reasons: 1) The occluded background objects will appear in the foreground due to the incomplete and sparse point clouds. 2) Some objects are wrongly obscured by noisy points. 3) Small camera movements will cause large replacements of points, especially for neighboring points. After obtaining multi-plane representations, our method learns multi-plane RGB images and corresponding blend weights with 3D convolution neural network, and the final output is generated by weighted summation. In order to improve the image quality, the 3 channel RGB is replaced as neural descriptors which also can be optimized during training. (More details in the full paper [5])

5.2 Results

In Fig. 5, we evaluate our approach on ScanNet [3] and Matterport 3D [1]. From the results, we can observe that our method can better handle occlusion relationships. For instance, the laptop behind the chair in the first row and the wall in the third row.

6 RENDERING AUGMENTATION

How to efficiently integrate rendering into data augmentation is worth exploring. Intuitively, rendering has many good properties, such as being physically reasonable, and diverse. Here, we take view augmentation for example.

A simple way to realize view augmentation is to estimate depth from the RGB image, and then change the camera pose to obtain novel view images via perspective projection where the missing areas are inpainted by Telea et al. [16]. Unfortunately, this approach will decrease the network performance. We suspect that inaccurate depth estimation and artifacts existing in the images result in degradation. Instead, the single image neural rendering is robust to depth estimation and can hallucinate the missing regions, but how to establish the correlations between novel view images and pixel-level annotations is still challenging.

In the future, more aspects (e.g. illumination, material) of rendering should be taken into consideration. What's more, current augmentation strategies mainly focus on static scenes, thus dynamic scene rendering seems to be a potential research direction.

7 CONCLUSION

In this paper, we introduce some findings in neural rendering and discuss its potential application. More specifically, the neural network can be used to imitate traditional rendering by providing essential components, and a multi-plane design benefits the novel view synthesis. At last, we discuss that neural rendering can be combined with downstream tasks. Consequently, we hope our findings can inspire more future works.

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